Melanocortin Model EQ

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$$\frac{d[\text{M_free}]}{dt} = k_\text{release_MMM} \cdot \text{M_MMM} \tag{1}$$
$$- k_\text{bind_MEGF8} \cdot \text{MEGF8} \cdot \text{M_free} \cdot (1 - f_\text{MB}) \tag{2}$$
$$- k_\text{bind_ATRN} \cdot \text{ATRN} \cdot \text{M_free} \tag{3}$$
$$+ k_\text{release_ATRN} \cdot \text{M_ATRN} \tag{4}$$
$$- k_\text{deg_MGRN1} \cdot \text{M_free} \tag{5}$$
$$+ k_\text{synth_MGRN1} \tag{6}$$

$$\frac{d[\text{M_MMM}]}{dt} = k_\text{bind_MEGF8} \cdot \text{MEGF8} \cdot \text{M_free} \cdot (1 - f_\text{MB}) \tag{7}$$
$$- k_\text{release_MMM} \cdot \text{M_MMM} \tag{8}$$
$$- k_\text{deg_MGRN1} \cdot \text{M_MMM} \tag{9}$$

$$\frac{d[\text{M_ATRN}]}{dt} = k_\text{bind_ATRN} \cdot \text{ATRN} \cdot \text{M_free} \tag{10}$$
$$- k_\text{release_ATRN} \cdot \text{M_ATRN} \tag{11}$$
$$- k_\text{deg_MGRN1} \cdot \text{M_ATRN} \tag{12}$$

$$\frac{d[\text{MC4R_s}]}{dt} = k_\text{synth_MC4R} \tag{13}$$
$$- k_\text{ub_MC4R} \cdot \text{M_ATRN} \cdot \text{MC4R_s} \tag{14}$$
$$- k_\text{deg_MC4R} \cdot \text{MC4R_s} \tag{15}$$

$$\frac{d[\text{MC4R_ub}]}{dt} = k_\text{ub_MC4R} \cdot \text{M_ATRN} \cdot \text{MC4R_s} \tag{16}$$
$$- k_\text{deg_ub_MC4R} \cdot \text{MC4R_ub} \tag{17}$$

$$f_\mathrm{MB} = \frac{\mathrm{MB}}{\mathrm{MB} + K_d, \mathrm{MB}} \tag{18}$$

Parameter	**Value**	**Units**	**Source**
$k_{\text{release_MMM}}$	0.01	s^{-1}	$K_d = 10$ nM $\times$ $k_{\text{bind_MEGF8}}$
$k_{\text{bind_MEGF8}}$	0.001	$\text{nM}^{-1}\text{s}^{-1}$	Diffusion limit, Berg & Purcell 1977
$k_{\text{bind_ATRN}}$	0.01	$\text{nM}^{-1}\text{s}^{-1}$	Diffusion limit, MASRPFA motif conserved
$k_{\text{release_ATRN}}$	0.5	s^{-1}	$K_d = 50$ nM $\times$ $k_{\text{bind_ATRN}}$
$k_{\text{deg_MGRN1}}$	9.26×10^{-6}	s^{-1}	Mathieson 2018, $t_{1/2} = 20.8$ hr
$k_{\text{synth_MGRN1}}$	1.67×10^{-6}	nM s^{-1}	Steady state: $k_{\text{deg}} \times 0.18$ nM (PaxDb)
$k_{\text{ub_MC4R}}$	7.351×10^{-5}	$\text{nM}^{-1}\text{s}^{-1}$	Calibrated to 3x MC4R KO, Kong 2025 Fig 4B
$k_{\text{deg_MC4R}}$	2.57×10^{-6}	s^{-1}	3T3 surface protein $t_{1/2} > 75$ hr
$k_{\text{deg_ub_MC4R}}$	1.93×10^{-4}	s^{-1}	GPCR lysosomal $t_{1/2} \approx 1$ hr
$k_{\text{synth_MC4R}}$	4.63×10^{-7}	nM s^{-1}	Calibrated to 3x MC4R KO, Kong 2025 Fig 4B
$K_{d,\text{MB}}$	10.0	nM	NB270 nanobody benchmark
[MEGF8]	2.4	nM	PaxDb 0.04 ppm $\times$ 10x local factor
[ATRN]	39.4	nM	PaxDb 6.56 ppm
$[\text{MGRN1}]_{\text{total}}$	0.18	nM	PaxDb 0.03 ppm
$[MC4R_s]_0$	0.06	nM	PaxDb 0.01 ppm

Table 1: Model parameters with values, units, and literature sources.